

From SAS to Python: A Statistical Computing Migration

USDA | National Agricultural Statistics Services (NASS)

Hritika Chaturvedi, Data Fellow | *UCLA, B.S. in Computational and Systems Biology*

Anish Gupta, Data Fellow | *Rutgers University-New Brunswick, Master of Public Informatics*

Richard Keith, Data Fellow | *UCSB, B.S in Statistics and Data Science*

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Survey methodology, legacy modernization, cost reduction, cloud readiness, code translation

Summary:

This project translates the USDA NASS survey summary systems from SAS, a proprietary statistical language whose codebase dates to 1989, into Python. The summary system is the final processing stage that converts raw survey responses into official published estimates of U.S. agricultural production, acreage, and livestock counts at the state, regional, and national level. The Python replacement reproduces the statistical formulas associated with these categories. The result is a system that's cheaper and easier to maintain and deployable to modern cloud infrastructure.

FROM SAS TO PYTHON: A STATISTICAL COMPUTING MIGRATION

USDA NASS

Justin Van Wart (Mathematical Statistician), Marisa Reuber (Acting Director)

coding it forward >



Hritika Chaturvedi
Anish Gupta
Richard Keith

THE PROBLEM

WHAT HAPPENS AT NASS?

- NASS conducts agricultural surveys (crop acreage, production, livestock counts, etc.) across all 50 states. After responses come in, their summary system turns raw responses into official published estimates.
 - **Collect:** Survey responses arrive from farmers
 - **Merge:** Combine responses with the sample design
 - **Weight:** Apply statistical weights so the sample represents the full population (scaling up a poll of 1,000 to represent an entire state)
 - **Summarize:** Compute totals, averages, ratios, and margins of error
 - **Aggregate:** Roll state results up to regional and national levels

WHY CHANGE A SYSTEM THAT WORKS?

- The codebase was written in 1989: hard to modify; few people still fluent in it
- SAS is proprietary software: expensive licensing
- Cannot integrate with modern web services or cloud
- Knowledge concentrated in few staff; institutional risk
- Difficult to audit what changed and when

THE GOAL

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- Produce identical statistical results using Python, an open-source language with a large talent pool, while maintaining:
 - Automated testing: Catch errors before they reach production
 - Web API access: Other systems can request summaries on demand
 - Cloud deployment: Scale across surveys without dedicated servers
 - Version control (Git): Full audit trail of every change

METHODS

TRANSLATE FIRST, MODERNIZE SECOND

- We translated the exact formulas from SAS into Python, then verified that both systems produce the same numbers:
 - Replaced hard coded survey names in code with configuration files (JSON) — run any survey without editing code
 - Replaced SAS-only output formats with standard CSV files — readable by Excel, R, Python, or any tool

HOW WE VALIDATED CORRECTNESS

- Ensuring the New System Matches the Old:
 - Side-by-side comparison: Run SAS and Python on identical inputs; compare every output number
- Acceptance Criteria:
 - Relative difference in totals: $< 5\%$
 - Relative difference in standard errors: $< 5\%$
 - Output file structure matches SAS column-for-column

ANALYSIS

SUMMARIZING A FRUIT SURVEY

Table A. Synthetic input records (sample of 12 from 1,200 total). Weight = farms represented; Usable = complete enough to include.

State	Farmer ID	Stratum	Bearing Acres	Production (tons)	Weight	Usable?
California	F-0012	Large	320	210	4.2	Yes
California	F-0045	Medium	85	52	6.1	Yes
California	F-0078	Small	12	7	15.3	Yes
Washington	F-0103	Large	140	95	3.8	Yes
Washington	F-0156	Medium	45	28	7.4	No
Washington	F-0189	Small	8	5	18.0	Yes
Idaho	F-0201	Large	95	60	5.0	Yes
Idaho	F-0234	Medium	30	18	9.2	Yes
Idaho	F-0267	Small	5	3	22.1	Yes
Oregon	F-0301	Large	110	72	4.5	Yes
Oregon	F-0334	Medium	40	24	8.0	Yes
Oregon	F-0367	Small	6	4	19.7	No

Illustrative synthetic data. No real USDA survey responses or official statistics are shown.

WHAT THE SUMMARY PIPELINE PRODUCES

Table B. Synthetic state summary output. CV = coefficient of variation; lower means more precise.

State	Variable	Estimated Total	Standard Error	CV (%)	Sample Size	Usable
California	Bearing Acres	142,350	8,420	5.9%	95	87
California	Production (tons)	89,200	6,100	6.8%	95	87
Washington	Bearing Acres	38,900	4,200	10.8%	50	42
Washington	Production (tons)	24,100	3,050	12.7%	50	42
Idaho	Bearing Acres	21,400	2,800	13.1%	35	33
Idaho	Production (tons)	13,500	1,900	14.1%	35	33
Oregon	Bearing Acres	29,700	3,400	11.4%	40	36
Oregon	Production (tons)	18,800	2,450	13.0%	40	36

STATE RESULTS COMBINE INTO REGIONAL TOTALS

Table C. Synthetic regional summary. Aggregation sums state estimates and combines variances statistically.

Region	Variable	Estimated Total	Standard Error	CV (%)	States Included
Pacific	Bearing Acres	211,950	10,200	4.8%	CA, WA, OR
Pacific	Production (tons)	132,100	7,300	5.5%	CA, WA, OR
Mountain	Bearing Acres	21,400	2,800	13.1%	ID
Mountain	Production (tons)	13,500	1,900	14.1%	ID
National	Bearing Acres	233,350	10,580	4.5%	All
National	Production (tons)	145,600	7,540	5.2%	All

SOME OUTPUTS ARE RATIOS

Table D. Synthetic ratio output. Ratio variances use special formulas because both parts are estimated from the sample.

State	Ratio Variable	Numerator Total	Denominator Total	Ratio Estimate	SE of Ratio	CV (%)
California	Yield (tons/acre)	89,200	142,350	0.627	0.041	6.5%
Washington	Yield (tons/acre)	24,100	38,900	0.620	0.068	11.0%
Idaho	Yield (tons/acre)	13,500	21,400	0.631	0.079	12.5%
Oregon	Yield (tons/acre)	18,800	29,700	0.633	0.072	11.4%

PYTHON VS. SAS COMPARISON

Table E. Side-by-side validation on synthetic data. Exact numerical parity confirms the translation is correct.

State	Variable		SAS Result	Python Result		Difference	Relative Diff
California	Bearing Acres Total		142,350.42	142,350.42		0.00	0.000%
California	Production SE		6,100.18	6,100.18		0.00	0.000%
Washington	Bearing Acres CV		10.793%	10.793%		0.000%	0.000%
Idaho	Yield Ratio		0.63084	0.63084		0.00000	0.000%
Oregon	Production Total		18,801.55	18,801.55		0.00	0.000%

CONCLUSION

INSIGHTS & REFLECTIONS

- What worked well:
 - Simply getting started before improving/redesigning
 - Documenting decisions; tracking progress through small milestones
 - Working closely with our supervisor; knowledge-sharing across teammates
 - Making code changes, reviewing, debugging, repeat — smooth machine
- Challenges:
 - Some SAS formulas had minimal documentation, required cross-referencing documents and consulting staff
 - SAS converts between text and numbers, leading to specific bugs
 - Minimizing mismatches with data that is constantly changing

THANK YOU

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